# Multimodal Deepfake Detection Using Audio-Visual Fusion Networks

**A Project Report Submitted in Fulfillment of the Requirements for the Degree of**

**Bachelor of Technology**

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# CERTIFICATE

This is to certify that the project report titled **"Multimodal Deepfake Detection Using Audio-Visual Fusion Networks"** submitted by **Divyanshu Ranjan, Sakshi Priya, Shubham Raj** in fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering (Data Science)** during the academic year 2025 - 26, is a record of bonafide work carried out under my supervision.

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# DECLARATION

We hereby declare that the project work entitled **"Multimodal Deepfake Detection Using Audio-Visual Fusion Networks"** carried out at the Department of Computer Science and Engineering (Data Science), Haldia Institute of Technology, is an authentic record of our own work under the supervision of Ms. Dipannita Basu. This report has not been submitted for any other degree or diploma elsewhere.

**Divyanshu Ranjan**
**Sakshi Priya**
**Shubham Raj**

# ACKNOWLEDGEMENT

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**Divyanshu Ranjan**

**Sakshi Priya**

**Shubham Raj**

# ABSTRACT


Deepfake technology has advanced rapidly in recent years, making manipulated images, videos, and audio recordings increasingly realistic and difficult to identify manually. While these technologies have useful applications, they also introduce serious concerns related to misinformation, identity misuse, and digital media manipulation. This project presents a Multimodal Deepfake Detection System that combines image, video, and audio analysis to improve the reliability and robustness of deepfake detection.

The image-based module utilizes EfficientNet-B4 and facial landmark analysis to identify spatial and geometric inconsistencies in manipulated images. The video-based module performs frame extraction, face alignment, and temporal feature analysis to detect motion-related artifacts in manipulated videos. The audio-based module uses Log-Mel spectrogram representations and deep learning techniques to identify spoofed or synthetic speech signals.

The extracted features from all three modalities are integrated using feature-level fusion to generate a unified representation for final classification. By combining visual, temporal, and acoustic information, the proposed framework improves the overall capability of detecting manipulated multimedia content compared to single-modality approaches.

The proposed system follows a modular and scalable architecture that supports future enhancement and deployment. The proposed framework also includes a graphical user interface for real-time multimedia authenticity verification and prediction visualization. The framework aims to provide a reliable solution for multimedia authenticity verification and contribute toward digital media forensics and deepfake detection research.

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# Chapter 1

# Introduction

## 1.1 Background

Recent advancements in Artificial Intelligence (AI) and Deep Learning (DL) have made it possible to generate highly realistic synthetic media, commonly known as deepfakes. These manipulated images, videos, and audio recordings are created using advanced techniques such as Generative Adversarial Networks (GANs), autoencoders, and voice cloning models. Although deepfake technology has useful applications in entertainment, media production, and accessibility, its misuse has raised serious concerns regarding misinformation, identity theft, digital fraud, and media manipulation.

Modern deepfakes have become increasingly realistic, making it difficult for humans and traditional detection systems to differentiate between authentic and manipulated content. Earlier deepfake detection approaches mainly focused on identifying visual artifacts or audio inconsistencies using a single modality. However, recent deepfake generation techniques are capable of minimizing such artifacts, reducing the effectiveness of single-modality detection systems.

To address this challenge, multimodal deepfake detection has emerged as a more reliable approach. By analyzing facial images, video frames, and audio signals together, multimodal systems can identify inconsistencies across different modalities that are difficult to manipulate simultaneously. This project focuses on developing a multimodal deepfake detection framework capable of analyzing image, video, and audio data for improved detection accuracy and robustness.

## 1.2 Aim of the Project

The aim of this project is to design and implement a **multimodal deepfake detection system** capable of analyzing manipulated images, videos, and audio recordings using deep learning and feature fusion techniques. The framework integrates image-based, video-based, and audio-based detection modules to improve the reliability and robustness of deepfake detection.

## 1.3 Objectives

The main objectives of the project are as follows:

1. To study existing deepfake generation and detection techniques through literature review.
2. To preprocess and prepare datasets for image, video, and audio modalities.
3. To develop an image-based deepfake detection module using EfficientNet-B4 feature extraction and facial landmark analysis.
4. To develop a video-based detection module using frame extraction, face alignment, and temporal feature analysis.
5. To develop an audio-based deepfake detection module using spectrogram-based speech analysis and deep learning techniques.
6. To implement a multimodal feature fusion framework that combines image, video, and audio representations for final classification.
7. To evaluate system performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis.
8. To identify existing challenges and propose improvements for future work.

## 1.4 Problem Statement

The rapid improvement in deepfake generation techniques has made manipulated media increasingly difficult to detect using traditional methods. Most existing detection systems rely on single-modality detection approaches such as image-only or audio-only analysis, which often fail against highly realistic deepfakes.

Modern deepfakes can simultaneously manipulate facial appearance, voice, and video content, making standalone detection systems less reliable. In addition, challenges such as temporal inconsistencies, advanced voice cloning, and poor cross-dataset generalization further reduce detection accuracy.

Therefore, there is a need for a more robust detection framework capable of analyzing multiple modalities together. This project addresses these challenges by developing a multimodal deepfake detection system that combines visual, temporal, and acoustic information to improve the reliability and effectiveness of manipulated media detection.

# Chapter 2

# Literature Review

## 2.1 Introduction

The rapid advancement of deep learning techniques has significantly improved the realism of synthetic media, leading to growing concerns regarding the misuse of deepfake technology. Large-scale datasets and benchmark challenges have accelerated research in this field [1]. Researchers have proposed various approaches to detect manipulated images, videos, and audio content. However, each modality presents unique challenges due to varying manipulation techniques and data characteristics.

This chapter reviews existing detection methods with a focus on image-based, video-based, audio-based, and multimodal approaches. The strengths and limitations of these methods are discussed to highlight the motivation behind developing a multimodal deepfake detection framework.

## 2.2 Image-Based Deepfake Detection

Image-based detection techniques mainly focus on identifying spatial artifacts and inconsistencies introduced during face manipulation. One of the most significant contributions in this domain is the FaceForensics++ dataset introduced by Rössler et al. [2]. This dataset enabled large-scale evaluation of detection models and demonstrated that Convolutional Neural Networks (CNNs) can effectively learn manipulation artifacts from facial images.

Afchar et al. [3] proposed MesoNet, a compact CNN architecture designed to detect facial forgeries by analyzing mesoscopic-level features rather than high-level semantic information. The lightweight design of MesoNet makes it suitable for real-time applications; however, its performance decreases when applied to high-quality deepfakes with minimal visible artifacts.

Li et al. [4] introduced the Face X-Ray technique, which detects blending artifacts between real and manipulated facial regions. Other approaches have explored camera model fingerprints to identify inconsistencies introduced during manipulation, such as the Noiseprint method proposed by Cozzolino and Verdoliva [5].

Although image-based approaches are effective for static analysis, they struggle against highly realistic deepfakes and cannot capture temporal or cross-modal inconsistencies. This limitation motivates the inclusion of additional modalities such as video and audio.

## 2.3 Video-Based Deepfake Detection

Video-based approaches extend image analysis by incorporating temporal information across consecutive frames. These methods analyze inconsistencies in facial movements, eye blinking patterns, head pose dynamics, and frame-to-frame coherence. Early studies demonstrated that biological signals such as abnormal eye blinking can reveal manipulated videos [6].

Techniques such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and two-stream neural architectures have been widely adopted to model temporal dependencies in video sequences [7]. Several studies have shown that temporal artifacts, including unnatural facial expressions and inconsistent motion patterns, can serve as strong indicators of video manipulation.

However, recent deepfake generation models are becoming increasingly capable of preserving temporal consistency, reducing the reliability of video-only systems. In addition, video-based methods often require large computational resources and extensive labeled datasets, making real-time deployment challenging.

These limitations highlight the need for complementary information from other modalities.

## 2.4 Audio Deepfake Detection

Audio deepfake detection focuses on identifying synthetic or spoofed speech generated using voice cloning and text-to-speech systems. The ASVspoof 2019 challenge introduced benchmark datasets and evaluation protocols for detecting spoofed audio [8]. Traditional approaches commonly rely on features such as Mel-Frequency Cepstral Coefficients (MFCCs), spectrogram representations, and phase-based acoustic features.

Recent research has explored deep learning and diffusion-based methods to improve the robustness and interpretability of audio spoof detection systems [9]. These approaches attempt to capture subtle acoustic patterns that differentiate genuine speech from synthetic speech.

Although audio-based methods perform effectively in controlled environments, their reliability decreases when audio quality is high or when synthetic speech is combined with realistic visual manipulation. This limits the effectiveness of standalone audio detection systems in practical scenarios.

## 2.5 Multimodal Deepfake Detection

To overcome the limitations of single-modality systems, researchers have proposed multimodal deepfake detection frameworks that integrate visual and audio information. These approaches exploit cross-modal inconsistencies, such as mismatches between lip movements and speech signals, to improve detection accuracy. Mittal et al. [10] demonstrated that combining affective audio-visual cues can significantly improve detection performance.

Multimodal systems generally employ feature-level fusion, where features from multiple modalities are combined into a unified representation, or decision-level fusion, where predictions from individual models are aggregated. By leveraging complementary information from different modalities, multimodal frameworks provide improved robustness against sophisticated deepfake attacks.

Despite their advantages, multimodal systems introduce challenges related to computational complexity, data synchronization, and implementation difficulty. Nevertheless, their improved reliability makes them a promising direction for deepfake detection research.

## 2.6 Research Gap

Although considerable progress has been made in deepfake detection, several research gaps still exist. Most existing systems rely on single-modality detection approaches, limiting their effectiveness against advanced deepfakes that exhibit high realism across visual and audio domains. In addition, many models suffer from poor generalization across datasets and manipulation techniques [11].

There is a need for a unified and scalable framework that integrates image, video, and audio analysis while maintaining robustness and computational efficiency. Addressing this gap forms the primary motivation for the proposed multimodal deepfake detection system.

## 2.7 Summary

This chapter reviewed existing approaches for image-based, video-based, audio-based, and multimodal deepfake detection. While single-modality systems provide useful insights, they are often insufficient against highly realistic manipulated media. Multimodal frameworks offer a more reliable solution by capturing inconsistencies across multiple modalities. These limitations motivate the development of the proposed multimodal deepfake detection framework discussed in the following chapters.

# Chapter 3

# System Design

## 3.1 Overview of the Proposed System

The proposed system is designed to detect manipulated media by combining information from image, video, and audio modalities. Unlike traditional deepfake detection approaches that rely on a single modality, the proposed framework performs independent analysis on each modality and later combines their learned representations through multimodal feature fusion. This improves the robustness of the system against highly realistic deepfake content.

The framework consists of three independent modules: image-based detection, video-based detection, and audio-based detection. Each module performs preprocessing, feature extraction, and classification using modality-specific techniques. The extracted embeddings are then normalized and fused into a unified multimodal representation for final classification.

The image-based module focuses on detecting spatial inconsistencies and geometric distortions in manipulated facial images. Deep visual features are extracted using EfficientNet-B4, while facial landmark analysis is performed using dlib. The video-based module analyzes temporal inconsistencies across video frames using aligned facial sequences and deep feature extraction techniques. Similarly, the audio-based module processes speech signals using spectrogram-based representations to identify spoofed or synthetic speech.

The modular architecture allows each modality-specific model to be trained and improved independently while maintaining compatibility with the overall framework. In addition, the system is being extended with a graphical user interface for future real-time inference and deployment.

## 3.2 System Architecture

The overall architecture of the proposed multimodal deepfake detection system is shown in Figure 3.1. The system follows a modular pipeline in which image, video, and audio data are processed separately before multimodal integration.

Initially, the input media is divided into its corresponding modalities. Facial images and video frames undergo preprocessing operations such as face detection, resizing, normalization, and alignment. Audio samples are resampled and converted into spectrogram representations suitable for feature extraction.

For image analysis, EfficientNet-B4 extracts deep visual embeddings, while dlib-based landmark extraction captures geometric facial structures. The video pipeline extracts frames and performs face alignment using MTCNN before temporal feature extraction. The audio pipeline converts speech signals into Log-Mel spectrograms and learns discriminative audio embeddings using deep learning models.

After modality-specific processing, the extracted embeddings are normalized and combined using feature-level fusion. The fused representation is then passed to the final classification layer, which predicts whether the media sample is real or fake.

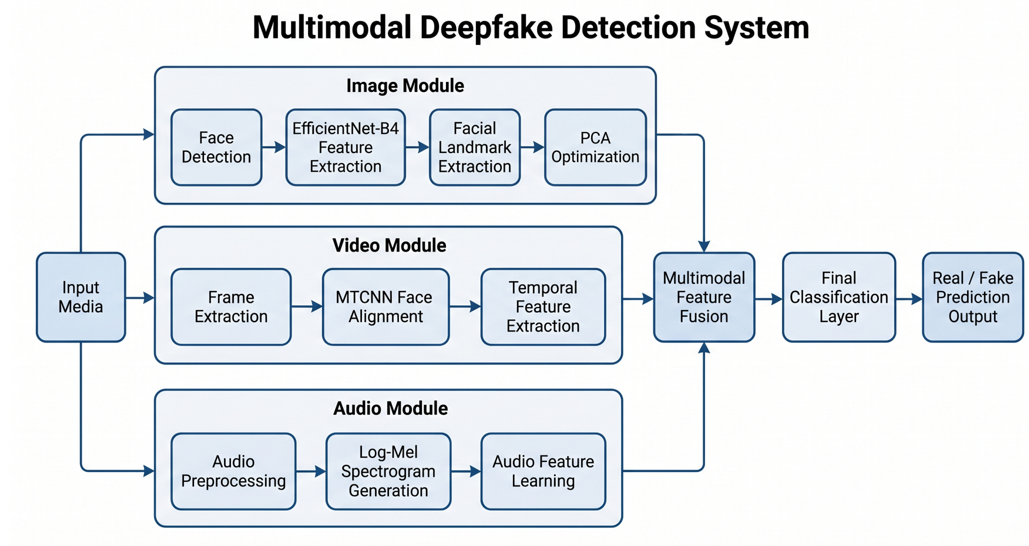


Figure 3.1: Overall architecture of the proposed multimodal deepfake detection system

## 3.3 Dataset Collection and Data Flow

The proposed framework utilizes publicly available benchmark datasets containing both authentic and manipulated media samples. Separate datasets are used for image, video, and audio modalities to ensure diversity in manipulation techniques and improve model generalization.

For image and video analysis, datasets such as FaceForensics++ and DFDC are used. The audio detection module utilizes spoofed and genuine speech samples from benchmark datasets such as ASVspoof.

Before training, the datasets undergo preprocessing operations such as resizing, normalization, frame extraction, audio resampling, and spectrogram generation. The processed data is then divided into training, validation, and testing subsets.

The overall data flow begins with multimodal media input. Each modality passes through its own preprocessing and feature extraction pipeline. The extracted embeddings are normalized and forwarded to the multimodal fusion module, where feature-level fusion is performed through concatenation. Finally, the fused feature vector is passed to the classification layer for real or fake prediction.

The complete data flow of the proposed framework is illustrated in Figure 3.2.

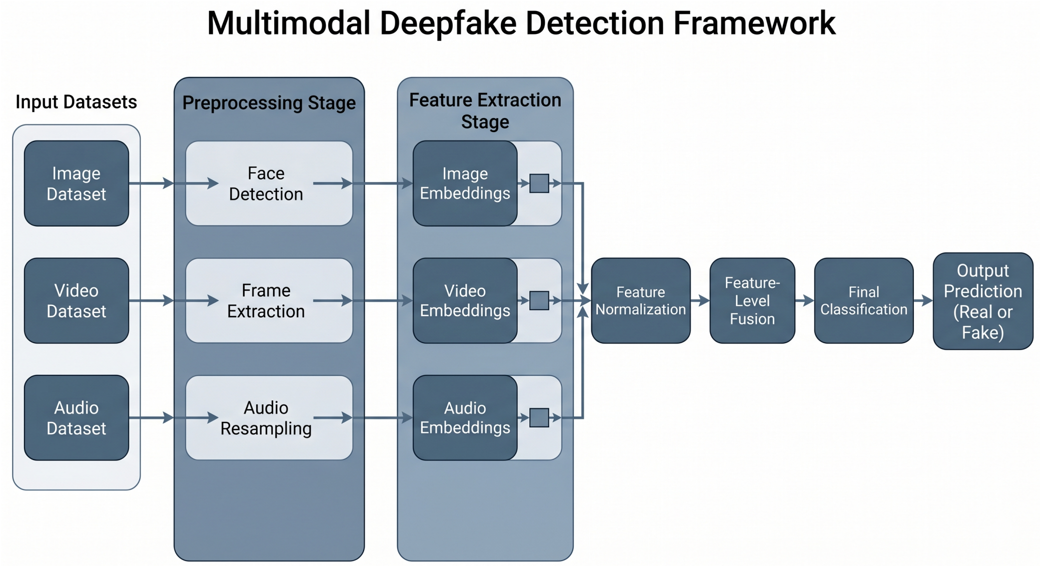


Figure 3.2: Data flow diagram of the proposed multimodal deepfake detection framework

## 3.4 Image-Based Deepfake Detection Module

The image-based deepfake detection module identifies spatial inconsistencies and geometric distortions introduced during facial manipulation. The module combines deep visual feature extraction with facial landmark analysis to improve detection robustness.

The image processing pipeline consists of face detection, preprocessing, deep feature extraction, landmark extraction, feature fusion, dimensionality reduction, and classification. The workflow of the image-based module is shown in Figure 3.3.

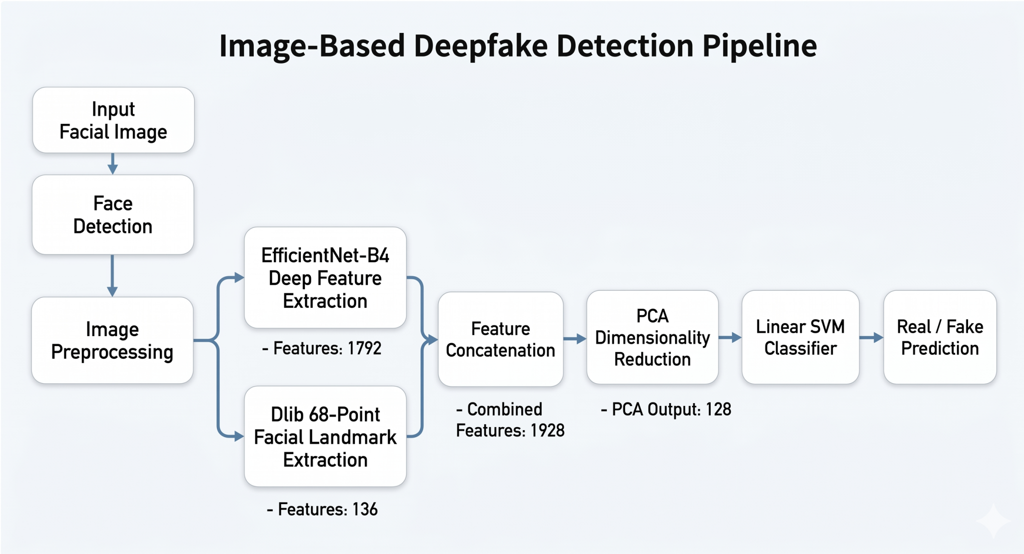


Figure 3.3: Workflow of the image-based deepfake detection module

### 3.4.1 Face Detection and Preprocessing

Facial regions are first detected, cropped, and resized to a fixed resolution suitable for feature extraction. Image normalization is applied to maintain consistency across samples. Data augmentation techniques such as horizontal flipping, rotation, and brightness adjustment are also used during training to improve generalization performance.

### 3.4.2 EfficientNet-B4 Feature Extraction

Deep visual features are extracted using the pretrained EfficientNet-B4 architecture. The model captures subtle texture inconsistencies, blending artifacts, and abnormal facial patterns introduced during manipulation. The extracted feature vector consists of 1792 dimensions.

### 3.4.3 Facial Landmark Extraction

Geometric facial information is extracted using the dlib 68-point facial landmark detector. The extracted landmarks capture facial structures such as eye positions, jawline structure, and mouth alignment. The landmark extraction process generates a 136-dimensional feature vector.

### 3.4.4 Feature Concatenation and PCA

The deep visual features and landmark features are combined through feature concatenation. Feature normalization is applied before dimensionality reduction using Principal Component Analysis (PCA). PCA reduces computational complexity while preserving important discriminative information.

### 3.4.5 Image Classification

The optimized feature vector is passed to a Linear Support Vector Machine (Linear SVM) classifier for real and fake prediction. The classifier generates the final prediction label along with a confidence score.

## 3.5 Video-Based Deepfake Detection Module

The video-based detection module identifies temporal inconsistencies and abnormal facial motion patterns present in manipulated videos. Unlike static image analysis, this module analyzes relationships between consecutive frames.

The video processing pipeline consists of frame extraction, face alignment, temporal feature extraction, embedding generation, and classification. The workflow is shown in Figure 3.4.

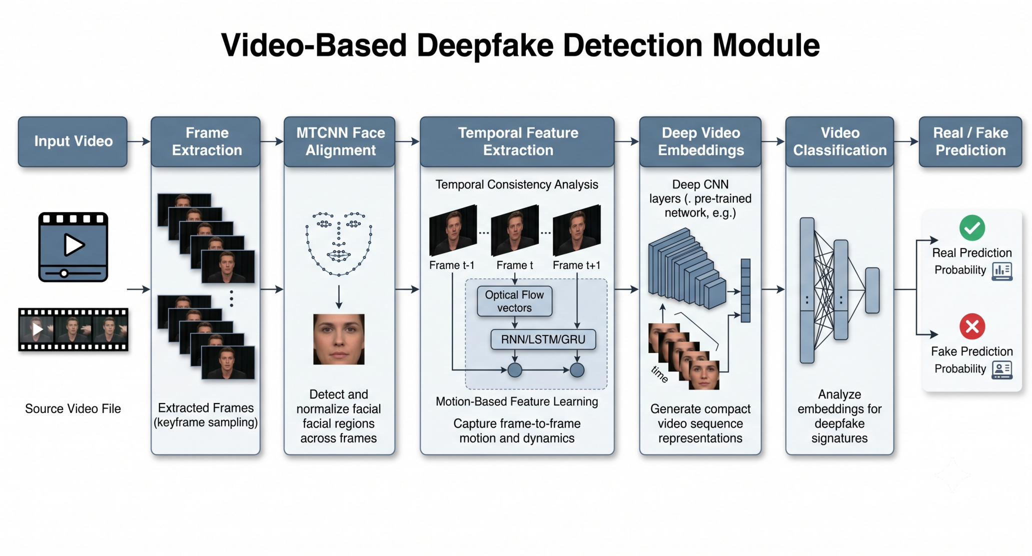


Figure 3.4: Workflow of the video-based deepfake detection module

### 3.5.1 Frame Extraction

Input videos are decomposed into individual frames at fixed intervals. Multiple frames are extracted to capture temporal information while maintaining computational efficiency.

### 3.5.2 MTCNN Face Alignment

Facial regions are detected and aligned using MTCNN. Face alignment improves consistency across extracted frames by correcting pose variations and spatial misalignment.

### 3.5.3 Temporal Feature Extraction

Temporal visual features are extracted using deep convolutional neural networks. The model captures inconsistencies such as abnormal facial motion, unnatural blinking, and frame transition artifacts.

Mixed precision training, learning rate scheduling, and adaptive optimization techniques are used to improve computational efficiency and convergence stability.

### 3.5.4 Video Classification

The extracted temporal embeddings are passed to the classification stage for real and fake prediction. The generated embeddings are also forwarded to the multimodal fusion module.

## 3.6 Audio-Based Deepfake Detection Module

The audio-based detection module identifies spoofed or synthetic speech generated using voice cloning and text-to-speech systems. The module analyzes spectral and acoustic characteristics of speech signals.

The audio pipeline consists of preprocessing, spectrogram generation, feature learning, embedding extraction, and classification. The workflow is shown in Figure 3.5.

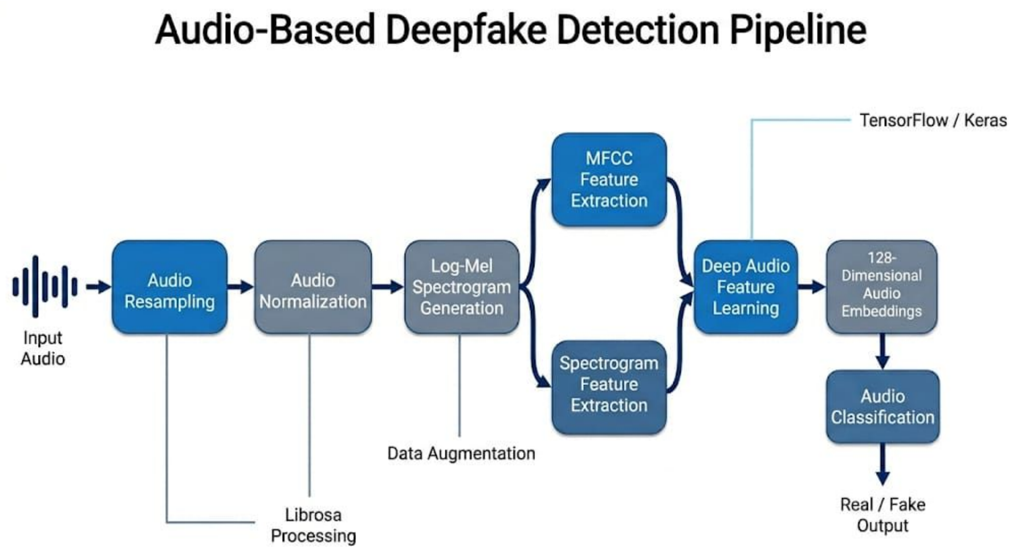


Figure 3.5: Workflow of the audio-based deepfake detection module

### 3.6.1 Audio Preprocessing

Audio samples are resampled and normalized to maintain consistency across all inputs. Noise reduction and trimming operations are applied to improve speech quality.

### 3.6.2 Spectrogram Generation

Speech signals are converted into Log-Mel spectrogram representations using the Librosa library. These spectrograms capture important acoustic characteristics required for spoof detection.

### 3.6.3 Audio Feature Learning

Deep learning models implemented using TensorFlow and Keras are used to learn discriminative audio embeddings. Data augmentation techniques are also applied during training to improve robustness.

### 3.6.4 Audio Classification

The extracted audio embeddings are passed to the classification stage for spoof detection. The generated embeddings are also utilized during multimodal feature fusion.

## 3.7 Multimodal Feature Fusion

The multimodal fusion module combines image, video, and audio embeddings to improve deepfake detection robustness. Since modern deepfake generation techniques attempt to minimize artifacts within individual modalities, combining multiple modalities improves overall reliability.

The workflow of the multimodal fusion framework is shown in Figure 3.6.

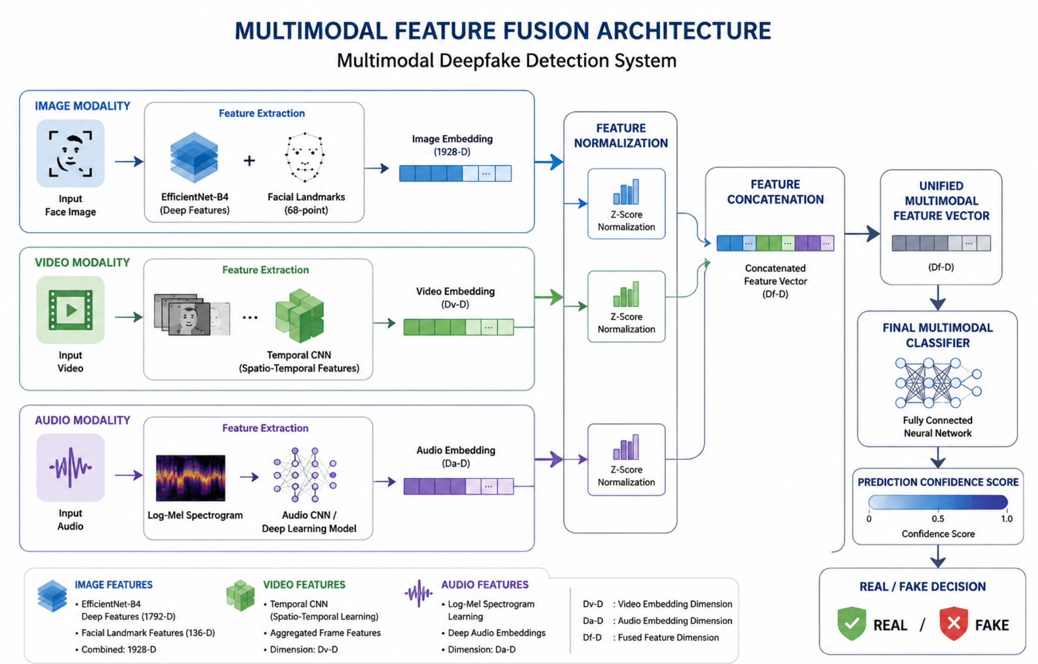


Figure 3.6: Feature-level multimodal fusion framework

### 3.7.1 Feature Normalization

Before fusion, the embeddings generated from different modalities are normalized independently to maintain balanced feature contribution and stabilize the learning process.

### 3.7.2 Feature-Level Fusion

The normalized embeddings are concatenated to generate a unified multimodal feature vector. The fused representation captures texture-based, temporal, and acoustic inconsistencies simultaneously.

### 3.7.3 Final Multimodal Classification

The fused multimodal representation is passed to the final classification layer for real and fake prediction. The classifier generates the final output label along with a confidence score.

## 3.8 User Interface and Deployment Architecture

A graphical user interface was developed to provide real-time interaction with the proposed multimodal deepfake detection framework. The interface allows users to upload image, video, and audio samples for authenticity verification through a unified platform.

The frontend interface was designed with a modern and responsive layout to improve usability and visualization of prediction results. Users can upload media samples directly through the interface, after which the corresponding modality-specific detection pipeline is executed automatically.

The system supports image, video, and audio analysis independently. During inference, the uploaded media passes through preprocessing and feature extraction stages before generating the final prediction. The interface displays prediction labels such as *Authentic Content* or *Deepfake Detected* along with confidence scores, processing time, and the model used for analysis.

Additional interface components such as recent analysis history, upload status indicators, and processing visualization were incorporated to improve user interaction and monitoring. The deployment architecture follows a modular design in which the frontend interface communicates with the backend multimodal detection pipeline.

The implemented interface establishes the foundation for future real-time deployment and scalable multimedia authenticity verification systems.

## 3.9 Overall Workflow of the System

The overall workflow of the proposed framework is summarized below:

1. Input media samples are collected from image, video, and audio datasets.
2. Modality-specific preprocessing operations are performed.
3. Deep feature extraction is carried out independently for each modality.
4. Image, video, and audio embeddings are generated.
5. The extracted embeddings are normalized and combined using feature-level fusion.
6. The fused representation is forwarded to the final classification stage.
7. The classifier predicts whether the input media is authentic or manipulated.

The complete workflow of the proposed system is shown in Figure 3.7.

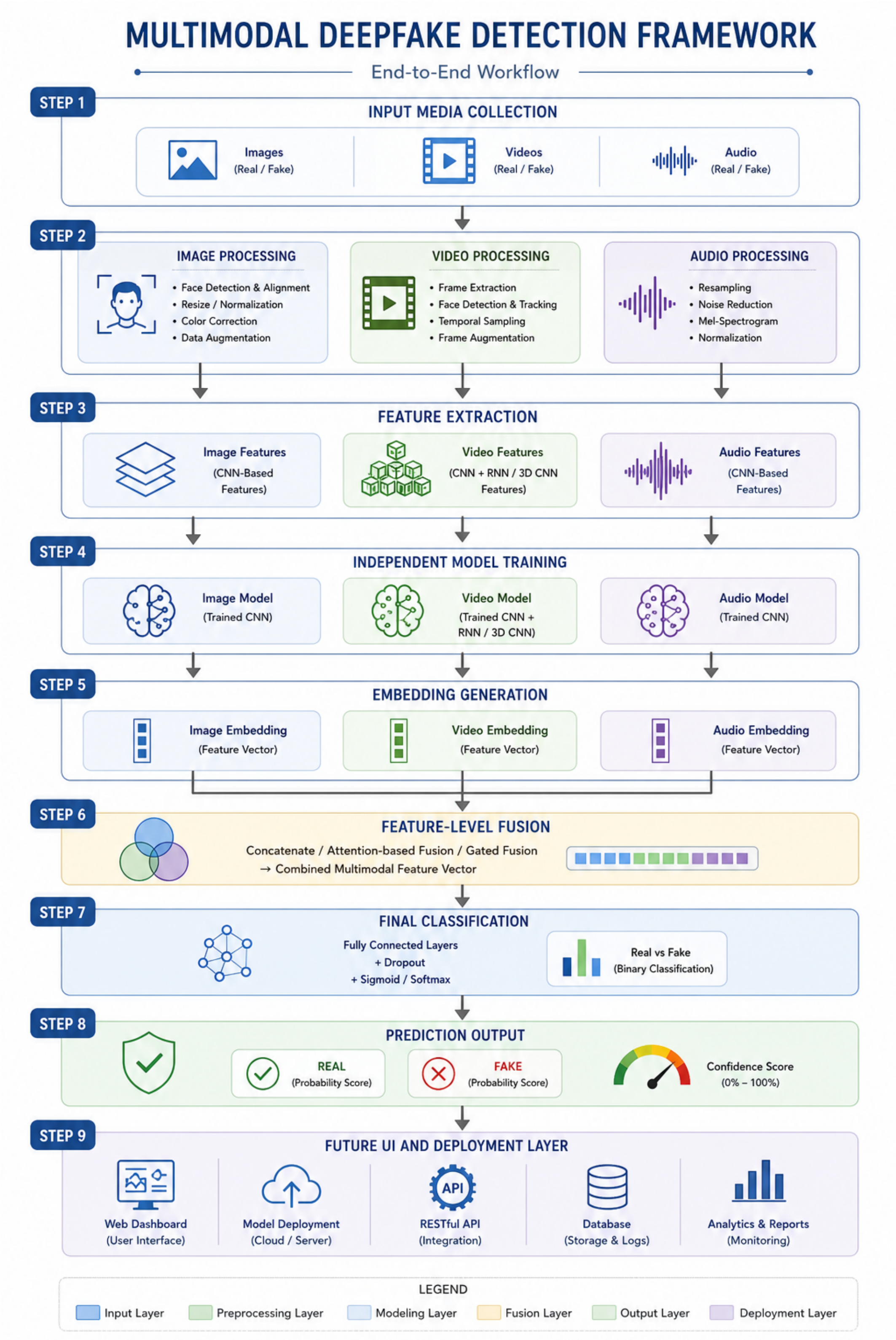


Figure 3.7: Overall workflow of the proposed multimodal deepfake detection system

## 3.10 Design Considerations

Several factors were considered during the design of the proposed framework. The first consideration was modularity, allowing each modality-specific pipeline to be trained and improved independently.

Another important consideration was robustness. By combining visual, temporal, and acoustic information, the framework becomes more resistant to sophisticated manipulation techniques.

Scalability and computational efficiency were also considered during system design. Techniques such as PCA, mixed precision training, and optimized preprocessing pipelines were incorporated to reduce computational overhead and improve training performance.

## 3.11 Summary

This chapter presented the system design of the proposed multimodal deepfake detection framework. The chapter discussed preprocessing operations, feature extraction techniques, modality-specific pipelines, multimodal fusion, and the overall workflow of the system.

The next chapter presents the implementation details of the proposed framework, including dataset preparation, model training, optimization strategies, and experimental setup.

# Chapter 4

# Implementation

## 4.1 Implementation Environment

The proposed multimodal deepfake detection framework was implemented using Python due to its extensive support for deep learning, computer vision, and audio processing libraries. GPU acceleration was utilized during training to improve computational efficiency and reduce processing time.

The image-based detection module was implemented using PyTorch and the `timm` library for EfficientNet-B4 feature extraction. Facial landmark detection and preprocessing operations were performed using OpenCV and dlib. The video-based module utilized OpenCV for frame extraction and MTCNN for face alignment.

The audio-based detection module was implemented using TensorFlow and Keras. Audio preprocessing and spectrogram generation were performed using the Librosa library. Feature normalization, PCA, and classification operations were implemented using Scikit-learn.

Mixed precision training techniques were incorporated within the video pipeline to improve GPU memory utilization and training performance.

Table 4.1 summarizes the software libraries used during implementation.

Table 4.1: Software Environment Used for Implementation

| **Component** | **Purpose** |
|---|---|
| Python | Core Programming Language |
| PyTorch | Deep Learning Implementation |
| TensorFlow / Keras | Audio Model Development |
| OpenCV | Image and Video Processing |
| dlib | Facial Landmark Detection |
| MTCNN | Face Alignment |
| Librosa | Audio Signal Processing |
| Scikit-learn | PCA and Classification |
| NumPy / Pandas | Data Processing |
| Matplotlib | Visualization |

Table 4.2 shows the hardware configuration used during experimentation.

Table 4.2: Hardware Configuration

| **Hardware Component** | **Specification** |
|---|---|
| Processor | Intel/AMD Multi-Core CPU |
| GPU | NVIDIA GPU with CUDA Support |
| RAM | 16 GB or Higher |
| Storage | SSD Storage |
| Operating System | Windows/Linux |

## 4.2 Dataset Preparation

The proposed framework utilizes publicly available datasets containing both authentic and manipulated media samples across image, video, and audio modalities.

For image and video analysis, datasets such as FaceForensics++ and DFDC were used. The audio-based module utilized benchmark spoofed speech datasets such as ASVspoof.

Before training, the datasets were organized into real and fake categories. Separate preprocessing pipelines were applied for image, video, and audio data.

For image preprocessing, facial regions were detected, cropped, and resized to the required input resolution. Data augmentation techniques such as flipping, rotation, and brightness adjustment were applied during training.

For video preprocessing, videos were decomposed into frames and facial regions were aligned using MTCNN. Audio preprocessing involved resampling, normalization, and spectrogram generation.

The processed datasets were divided into training, validation, and testing subsets to ensure unbiased evaluation.

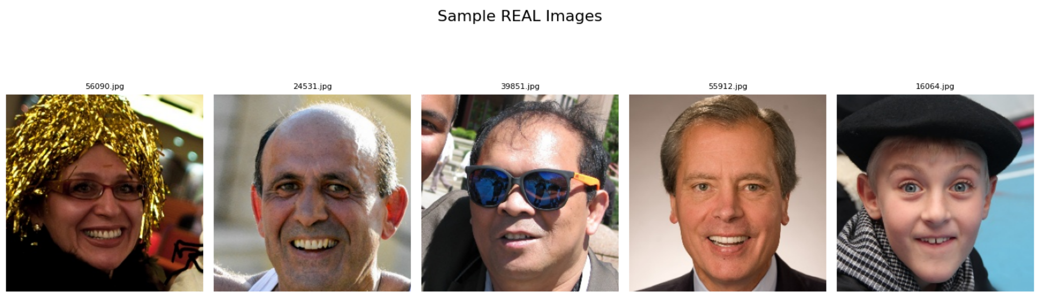


Figure 4.1: Sample authentic facial images used for training

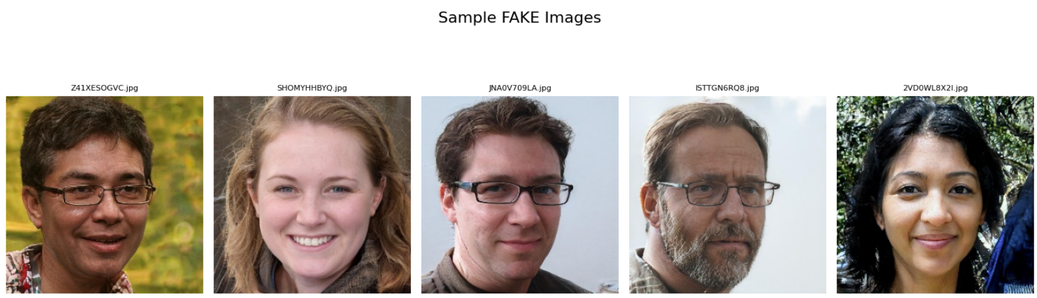


Figure 4.2: Sample manipulated facial images used for training

## 4.3 Image-Based Detection Implementation

The image-based detection module was implemented to identify spatial inconsistencies and geometric distortions present in manipulated facial images. The implementation combines deep visual feature extraction with facial landmark analysis.

The implementation pipeline consists of face detection, preprocessing, EfficientNet-B4 feature extraction, landmark extraction, PCA optimization, and classification.

### 4.3.1 Face Detection and Preprocessing

Facial regions were detected using OpenCV and dlib-based face detection techniques. The detected faces were cropped, resized, and normalized before feature extraction.

Data augmentation techniques such as horizontal flipping, random rotation, and brightness adjustment were applied during training to improve generalization.

### 4.3.2 EfficientNet-B4 Feature Extraction

Deep visual features were extracted using the pretrained EfficientNet-B4 architecture. The model captured texture inconsistencies, blending artifacts, and abnormal facial patterns commonly present in manipulated images.

The extracted embeddings generated a 1792-dimensional feature vector for each image.

### 4.3.3 Facial Landmark Extraction

Geometric facial information was extracted using the dlib 68-point facial landmark detector. The extracted landmarks captured important facial structures such as eye position, jawline structure, and mouth alignment.

The landmark extraction process generated a 136-dimensional feature vector.

### 4.3.4 Feature Fusion and PCA Optimization

The deep visual features and landmark features were combined through feature concatenation. PCA was then applied to reduce feature dimensionality and improve computational efficiency.

The combined feature vector initially contained 1928 dimensions and was reduced to 128 dimensions after PCA transformation.

### 4.3.5 Image Classification

The optimized feature vectors were passed to a Linear SVM classifier for real and fake prediction. The classifier generated the final prediction label along with a confidence score.

Table 4.3 summarizes the configuration of the image-based module.

Table 4.3: Configuration of the Image-Based Detection Module

| Parameter | Configuration |
| --- | --- |
| Deep Feature Extractor | EfficientNet-B4 |
| Input Resolution | $380 \times 380$ |
| Deep Feature Dimension | 1792 |
| Landmark Feature Dimension | 136 |
| Combined Feature Dimension | 1928 |
| Dimensionality Reduction | PCA |
| Reduced Feature Size | 128 |
| Classifier | Linear SVM |
| Framework | PyTorch + Scikit-learn |

## 4.4 Video-Based Detection Implementation

The video-based detection module was implemented to identify temporal inconsistencies and abnormal facial motion patterns present in manipulated videos.

The implementation pipeline consists of frame extraction, face alignment, temporal feature extraction, embedding generation, and classification. .

### 4.4.1 Frame Extraction

Input videos were decomposed into multiple frames using OpenCV. Frames were extracted at fixed intervals to maintain temporal consistency and reduce computational overhead.

### 4.4.2 Face Alignment using MTCNN

Facial regions from extracted frames were detected and aligned using MTCNN. Face alignment improved spatial consistency across all frames.


Figure 4.3: Sample extracted and aligned face crops from real and fake video frames

### 4.4.3 Temporal Feature Learning

Deep convolutional neural networks were used to learn temporal visual representations from aligned frames. The model captured inconsistencies such as abnormal facial motion and frame transition artifacts.

Mixed precision training and adaptive optimization techniques were used to improve training efficiency.

### 4.4.4 Video Classification

The extracted temporal embeddings were passed to the classification stage for real and fake prediction. The generated embeddings were also utilized during multimodal fusion.

Table 4.4 summarizes the video-based implementation settings.

Table 4.4: Configuration of the Video-Based Detection Module

| Parameter | Configuration |
|---|---|
| Frame Extraction Library | OpenCV |
| Face Alignment | MTCNN |
| Input Resolution | $224 \times 224$ |
| Optimizer | Adam |
| Learning Rate Scheduler | CosineAnnealingLR |
| Training Precision | Mixed Precision |
| Framework | PyTorch |
| Output Representation | Temporal Embeddings |

## 4.5 Audio-Based Detection Implementation

The audio-based detection module was implemented to identify spoofed and synthetic speech generated using voice cloning and text-to-speech systems.

The implementation pipeline consists of preprocessing, spectrogram generation, audio feature learning, embedding extraction, and classification.

### 4.5.1 Audio Preprocessing

Audio samples were resampled and normalized before feature extraction. Noise reduction and silence trimming operations were also applied where required.

### 4.5.2 Spectrogram Generation

Speech signals were converted into Log-Mel spectrogram representations using the Librosa library. These spectrograms were used as input for deep learning-based feature extraction.

### 4.5.3 Audio Feature Learning

Deep learning architectures implemented using TensorFlow and Keras were used to learn discriminative audio embeddings. Data augmentation techniques were also incorporated during training.

The audio feature learning network generated 128-dimensional embedding vectors for multimodal fusion.

### 4.5.4 Audio Classification

The extracted audio embeddings were passed to the classification stage for spoof detection. The output consisted of a prediction label and confidence score.

Table 4.5 summarizes the configuration of the audio-based module.

Table 4.5: Configuration of the Audio-Based Detection Module

| **Parameter** | **Configuration** |
|---|---|
| Audio Processing Library | Librosa |
| Feature Representation | Log-Mel Spectrogram |
| Embedding Dimension | 128 |
| Framework | TensorFlow / Keras |
| Optimizer | Adam |
| Learning Rate | $1 \times 10^{-4}$ |
| Batch Size | 32 |
| Epochs | 50 |
| Data Augmentation | Enabled |

## 4.6 Independent Model Training

Each modality-specific model was trained independently using its corresponding dataset and preprocessing pipeline. This allowed each module to learn modality-specific patterns without interference from other data sources.

The image model was trained using deep visual and landmark features, while the video model utilized aligned video frames and temporal embeddings. Similarly, the audio model was trained using spectrogram-based speech representations.

Training and validation losses were monitored continuously to evaluate convergence and avoid overfitting. Proper dataset separation was maintained throughout experimentation.

## 4.7 Multimodal Feature Fusion Implementation

After independent training, embeddings generated from image, video, and audio pipelines were integrated through feature-level fusion.

Initially, the embeddings from all modalities were normalized independently. The normalized embeddings were then concatenated to generate a unified multimodal feature vector.

The fused representation combines texture-based, temporal, and acoustic information for final classification.

## 4.8 Final Classification and Prediction

The unified multimodal representation generated after feature fusion was passed to the final classification stage for real and fake prediction.

The prediction pipeline generated a binary output along with a confidence score representing prediction certainty. The modular design also allows future integration of additional modalities and advanced fusion strategies.

## 4.9 User Interface Integration and Deployment

A graphical user interface was successfully integrated with the proposed multimodal deepfake detection framework to support real-time inference and user interaction.

The interface was developed to allow users to upload image, video, and audio samples directly through a unified platform. Depending on the uploaded media type, the corresponding modality-specific detection pipeline is executed automatically.

For image analysis, the interface displays the uploaded facial image along with the generated prediction result, confidence score, processing time, and model information. Similar interfaces were implemented for video and audio analysis, allowing users to analyze multimedia content independently.

The interface also includes additional features such as:

- Real-time upload and preprocessing visualization
- Confidence score display
- Prediction status indicators
- Recent analysis history
- Processing progress visualization
- Support for multiple media formats

The backend multimodal detection pipeline was integrated with the frontend interface through a modular deployment architecture. This design allows future scalability, independent model updates, and real-time deployment optimization.

Figures 4.4 to 4.10 illustrate the implemented graphical user interface and prediction workflow.

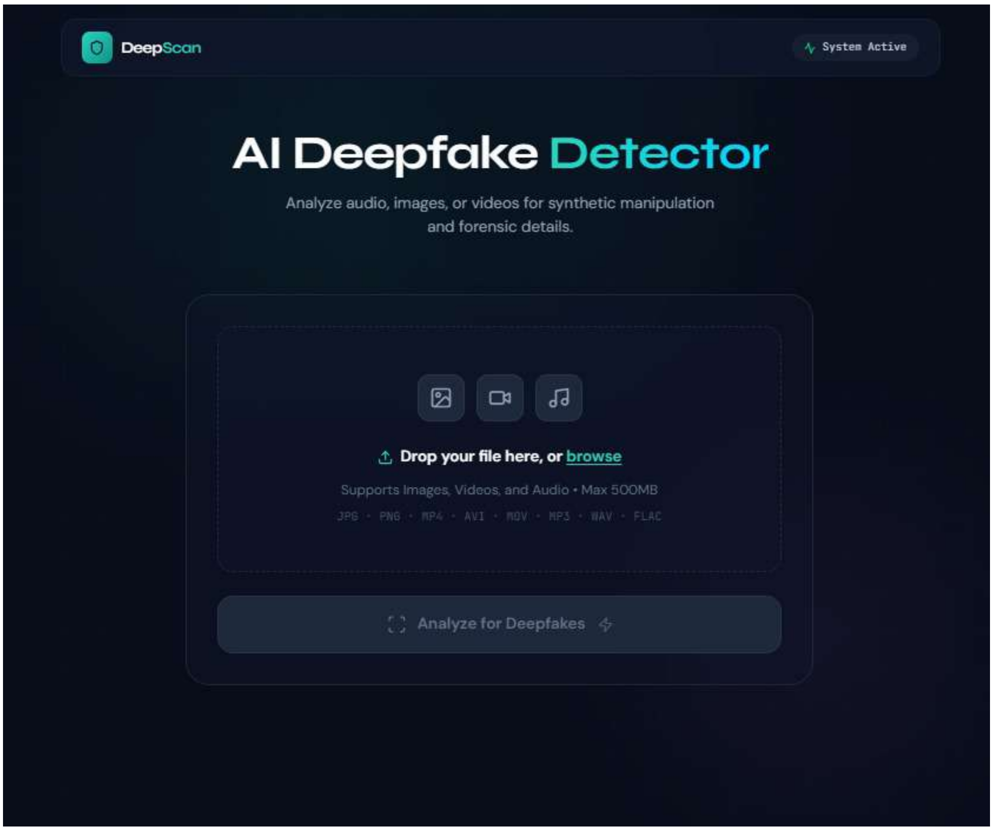


Figure 4.4: Homepage of the multimodal deepfake detection interface

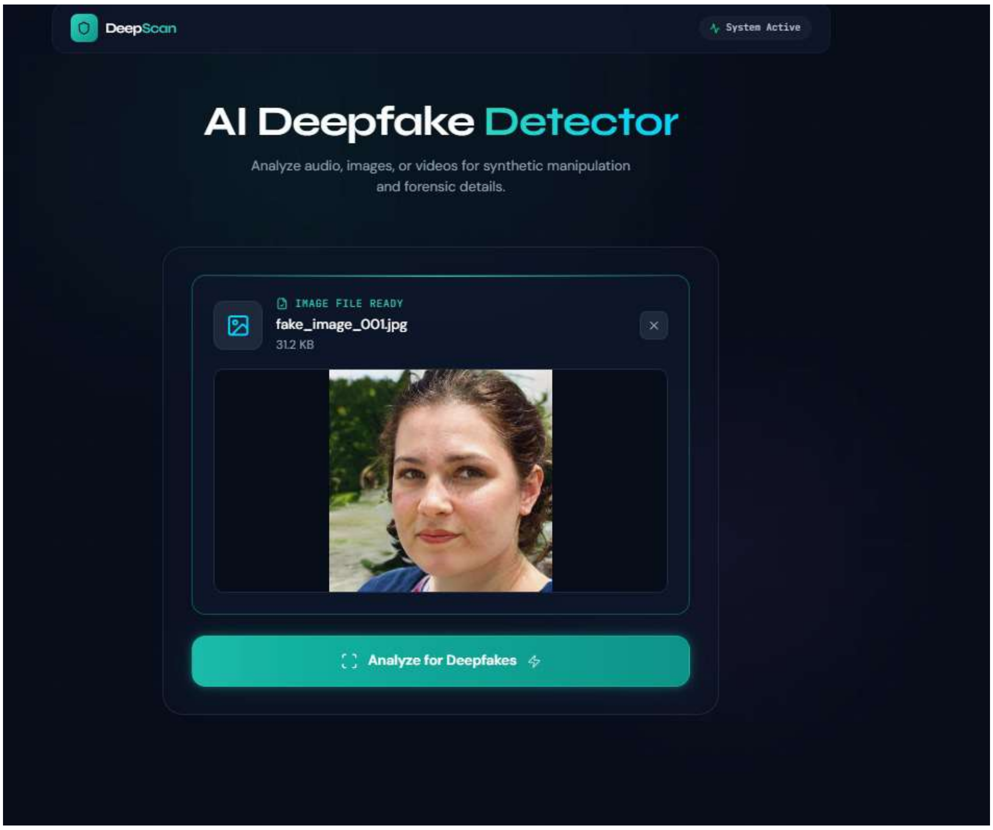


Figure 4.5: Image upload interface

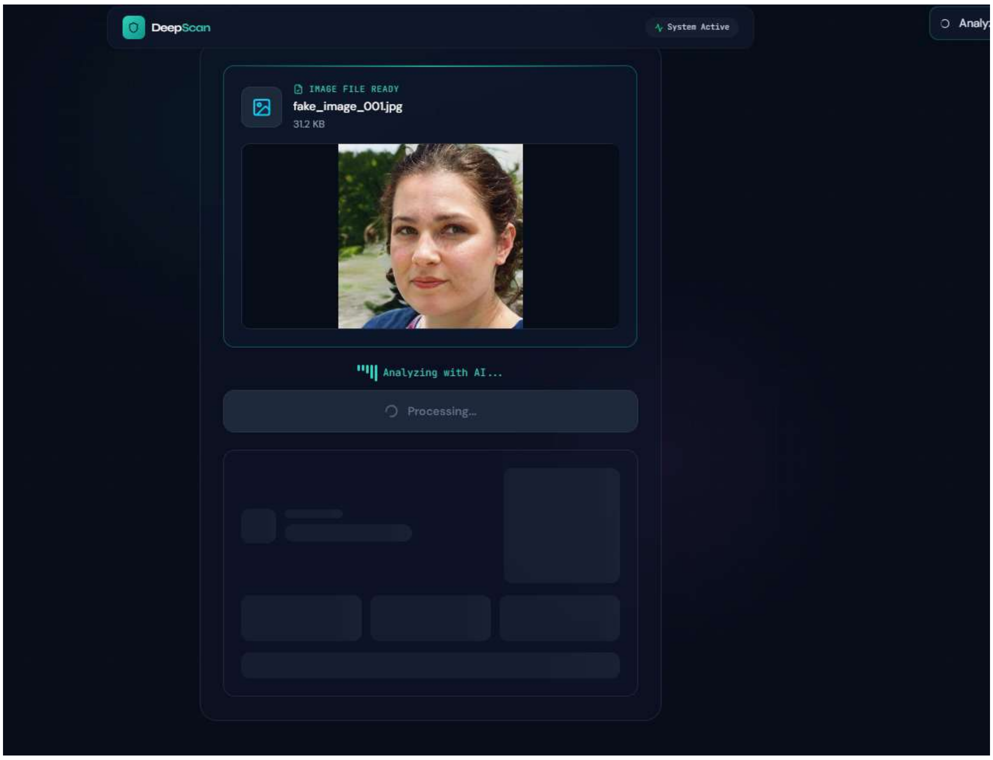


Figure 4.6: Image analysis processing interface

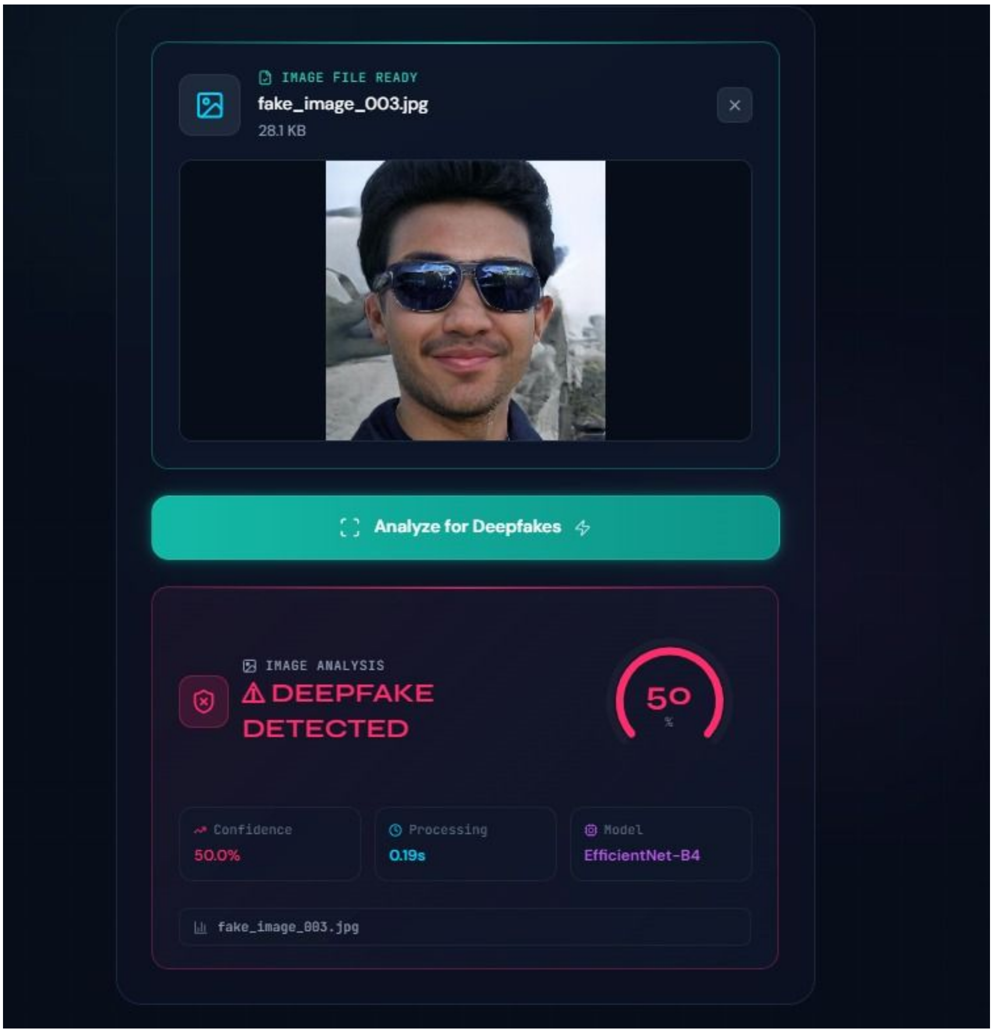


Figure 4.7: Deepfake image prediction result

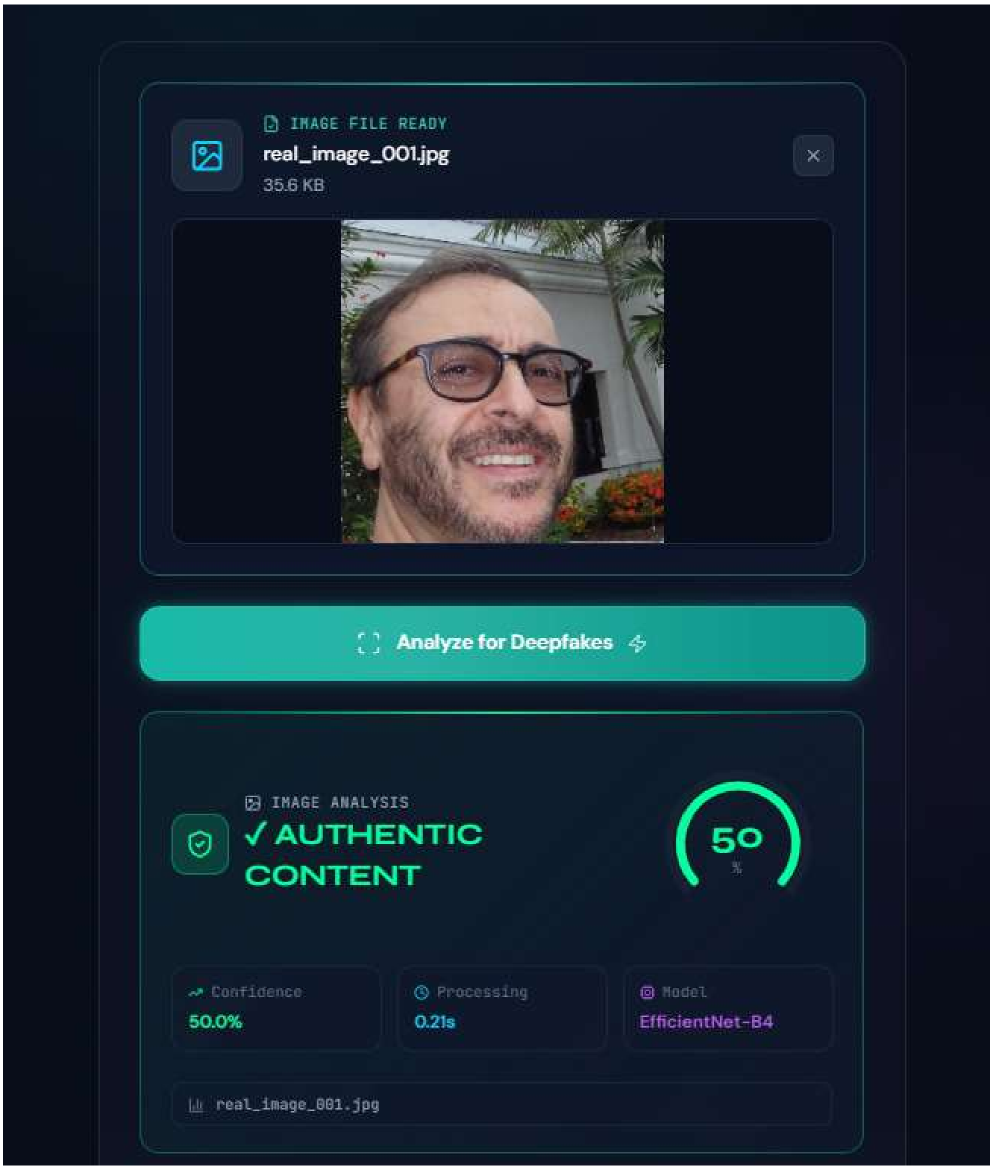


Figure 4.8: Authentic image prediction result

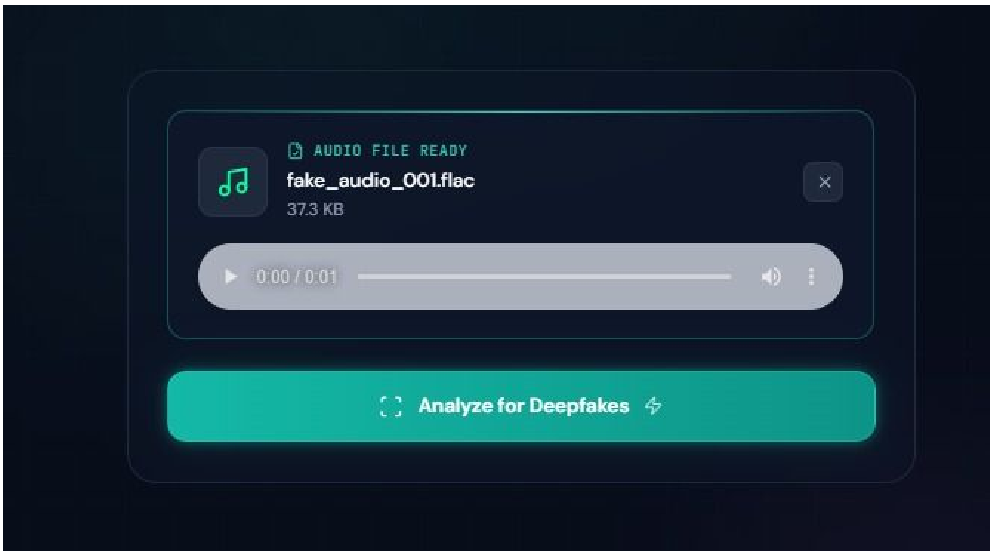


Figure 4.9: Audio analysis interface

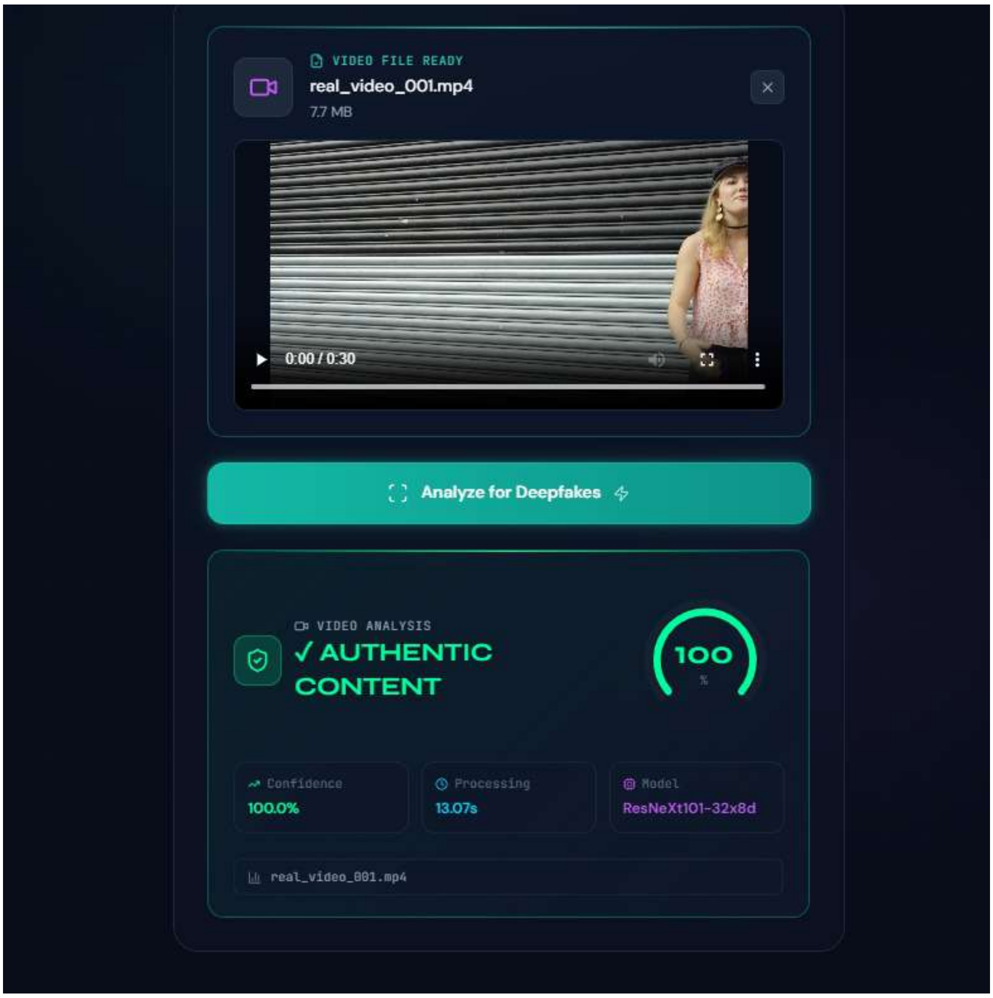


Figure 4.10: Video deepfake detection result

## 4.10 Summary

This chapter presented the implementation details of the proposed multimodal deepfake detection framework. The chapter discussed dataset preparation, modality-specific preprocessing, feature extraction, independent model training, and multimodal feature fusion.

The next chapter presents the experimental results and performance evaluation of the proposed framework.

# Chapter 5

# Results and Discussion

## 5.1 Experimental Setup

The proposed multimodal deepfake detection framework was evaluated using benchmark datasets containing authentic and manipulated image, video, and audio samples. Independent experiments were conducted for each modality-specific detection module before multimodal feature fusion.

The image-based detection module utilized EfficientNet-B4 for deep visual feature extraction and dlib-based facial landmark detection for geometric feature extraction. The extracted features were normalized, concatenated, and optimized using Principal Component Analysis (PCA) before classification using a Linear SVM classifier.

The video-based module utilized frame extraction, MTCNN face alignment, and temporal feature learning to identify inconsistencies across video sequences. Similarly, the audio-based module utilized Log-Mel spectrogram representations for spoofed speech detection.

The extracted embeddings from all modalities were later integrated using feature-level fusion for multimodal classification.

Performance evaluation was carried out using standard classification metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis.

## 5.2 Image-Based Detection Results

The image-based deepfake detection module was evaluated on a test dataset containing 20,000 facial images equally distributed between authentic and manipulated samples.

The proposed framework combines EfficientNet-B4 deep visual embeddings with dlib-based facial landmark features to capture both texture-based and geometric inconsistencies introduced during facial manipulation.

The model achieved an overall test accuracy of 86.59%, demonstrating effective classification performance for distinguishing authentic and manipulated facial images.

### 5.2.1 Performance Evaluation

Table 5.1: Performance Evaluation of the Image-Based Detection Model

| Metric | Value |
|---|---|
| Accuracy | 86.59% |
| Precision (REAL) | 0.88 |
| Recall (REAL) | 0.85 |
| F1-Score (REAL) | 0.86 |
| Precision (FAKE) | 0.86 |
| Recall (FAKE) | 0.88 |
| F1-Score (FAKE) | 0.87 |
| Macro Average F1-Score | 0.87 |

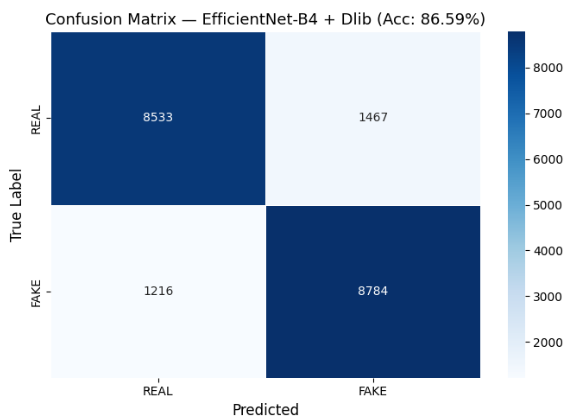


Figure 5.1: Confusion matrix of the image-based deepfake detection model

### 5.2.2 Discussion

The model correctly classified 8,533 authentic images and 8,784 manipulated images from the test dataset. The obtained results indicate that combining deep visual features with facial landmark information improves detection capability by capturing both texture-level artifacts and geometric inconsistencies.

Misclassifications mainly occurred in highly realistic deepfake samples where visual artifacts were minimal. The results also highlight the limitations of relying solely on static image analysis for advanced deepfake detection.

## 5.3 Video-Based Detection Results

The video-based deepfake detection module was evaluated using a test dataset containing 1,920 video samples equally distributed between authentic and manipulated videos.

The implemented framework utilizes temporal feature learning and face alignment to identify inconsistencies across consecutive video frames. Experimental evaluation showed that the video-based model achieved the highest performance among all individual modalities.

The model achieved an overall test accuracy of 93.02%, indicating strong capability in detecting manipulated video content.

### 5.3.1 Performance Evaluation

Table 5.2: Performance Evaluation of the Video-Based Detection Model

| **Metric** | **Value** |
|---|---|
| Accuracy | 93.02% |
| Precision (REAL) | 0.93 |
| Recall (REAL) | 0.93 |
| F1-Score (REAL) | 0.93 |
| Precision (FAKE) | 0.93 |
| Recall (FAKE) | 0.93 |
| F1-Score (FAKE) | 0.93 |
| Macro Average F1-Score | 0.93 |

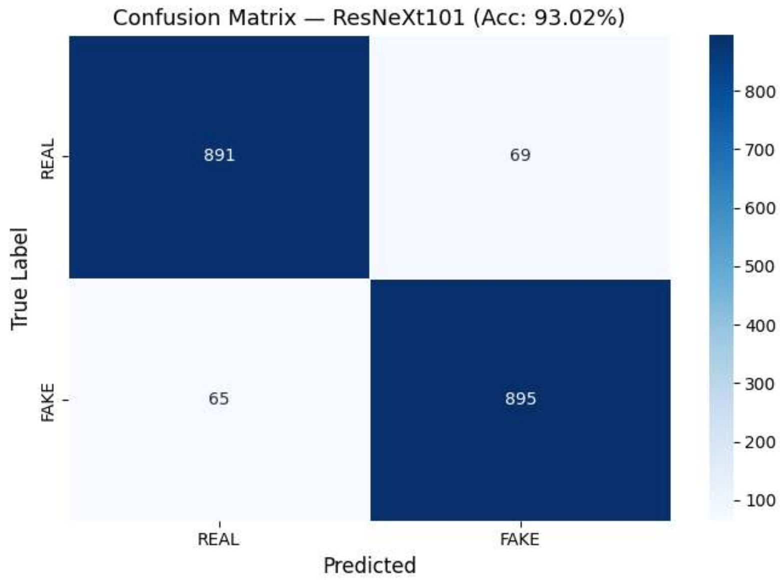


Figure 5.2: Confusion matrix of the video-based deepfake detection model

### 5.3.2 Discussion

The confusion matrix analysis shows that the model correctly classified 891 authentic videos and 895 manipulated videos. Only a small number of samples were misclassified during testing.

The superior performance of the video-based model highlights the importance of temporal information in deepfake detection. The model successfully captured inconsistencies such as abnormal facial motion, unnatural blinking behavior, and frame transition irregularities that are difficult to identify using static image analysis alone.

## 5.4 Audio-Based Detection Results

The audio-based deepfake detection module was evaluated using a balanced test dataset containing 14,710 speech samples consisting of both bonafide and spoofed audio recordings.

The model utilized Log-Mel spectrogram representations generated from preprocessed speech signals. Threshold optimization was performed using validation data, and the best performance was obtained at a decision threshold of 0.60.

Experimental evaluation showed that the audio-based detection module achieved an overall accuracy of 87.59% with a macro F1-score of 87.58%.

### 5.4.1 Performance Evaluation

Table 5.3: Performance Evaluation of the Audio-Based Detection Model

| **Metric** | **Value** |
|---|---|
| Accuracy | 87.59% |
| Macro F1-Score | 87.58% |
| Precision (Bonafide) | 0.85 |
| Recall (Bonafide) | 0.91 |
| F1-Score (Bonafide) | 0.88 |
| Precision (Spoof) | 0.91 |
| Recall (Spoof) | 0.84 |
| F1-Score (Spoof) | 0.87 |
| Best Threshold | 0.60 |

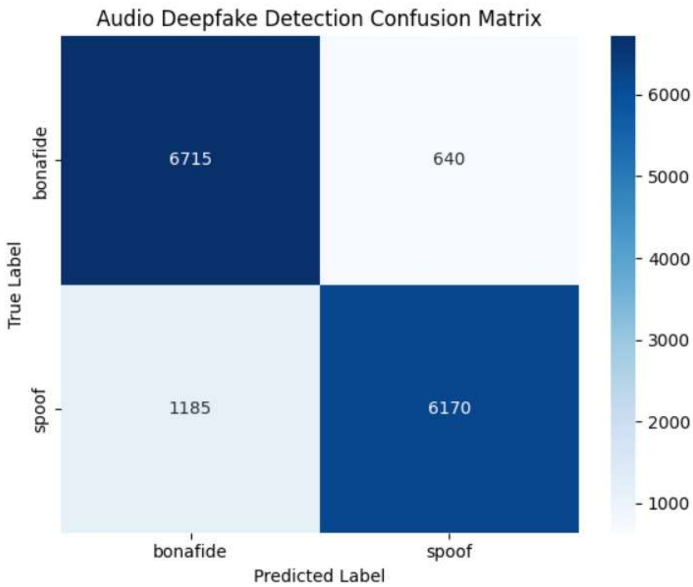


Figure 5.3: Confusion matrix of the audio-based deepfake detection model

### 5.4.2 Discussion

The model correctly classified 6,715 bonafide speech samples and 6,170 spoofed speech samples. The results indicate that acoustic artifacts introduced during synthetic speech generation remain detectable despite improvements in modern voice cloning techniques.

The audio-based model demonstrated balanced classification performance across both classes and provided complementary information for multimodal deepfake detection.

## 5.5 Multimodal Fusion Results

The multimodal fusion framework integrates image, video, and audio embeddings through feature-level fusion to improve deepfake detection robustness and reliability.

The extracted embeddings from all three modalities were normalized and concatenated into a unified multimodal feature representation before final classification.

Experimental evaluation of the multimodal framework achieved an overall accuracy of 83% on the test dataset.

### 5.5.1 Performance Evaluation

Table 5.4: Performance Evaluation of the Multimodal Fusion Model

| **Metric** | **Value** |
|---|---|
| Accuracy | 83% |
| Precision (REAL) | 0.84 |
| Recall (REAL) | 0.83 |
| F1-Score (REAL) | 0.83 |
| Precision (FAKE) | 0.83 |
| Recall (FAKE) | 0.84 |
| F1-Score (FAKE) | 0.84 |
| Macro Average F1-Score | 0.83 |

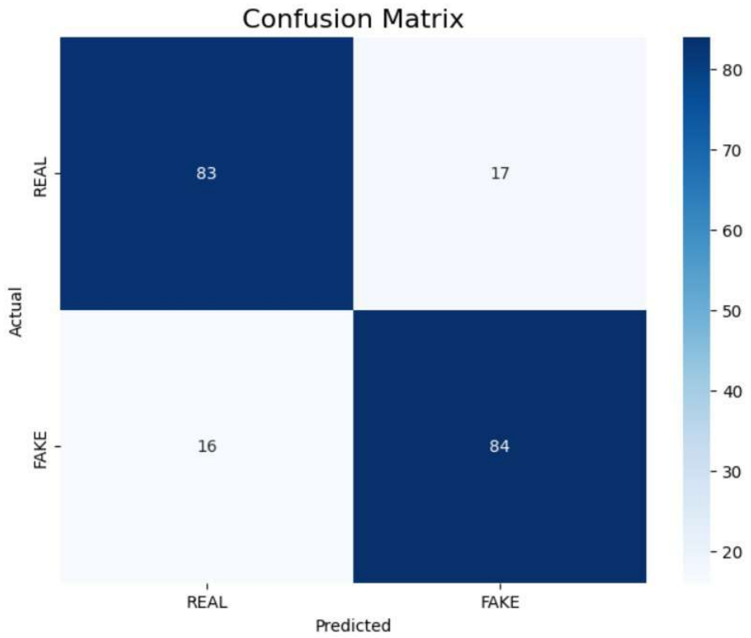


Figure 5.4: Confusion matrix of the multimodal deepfake detection model

### 5.5.2 Discussion

The multimodal framework demonstrated stable classification performance by integrating visual, temporal, and acoustic representations into a unified detection pipeline.

Although the multimodal accuracy was lower than the standalone video-based model, the integrated framework provides improved robustness and flexibility for handling different media modalities within a single system architecture.

The obtained results also indicate that multimodal feature fusion remains sensitive to embedding quality, dataset synchronization, and feature balancing across modalities. Further optimization of fusion strategies and larger synchronized datasets may improve overall multimodal performance.

## 5.6 Comparative Analysis of Modality-Specific Models

Table 5.5 compares the performance of the image-based, video-based, audio-based, and multimodal detection frameworks.

Table 5.5: Comparative Performance of Deepfake Detection Models

| Detection Module | Accuracy | Macro F1-Score |
|---|---|---|
| Image-Based Detection | 86.59% | 0.87 |
| Video-Based Detection | 93.02% | 0.93 |
| Audio-Based Detection | 87.59% | 0.88 |
| Multimodal Fusion | 83% | 0.83 |

The comparative analysis shows that the video-based detection module achieved the highest overall performance due to its ability to capture temporal inconsistencies across video frames.

The image-based and audio-based models also demonstrated strong classification capability for manipulated media detection. The multimodal framework provided a unified architecture for integrating information from multiple modalities, although additional optimization is required to further improve fusion performance.

## 5.7 Summary

This chapter presented the experimental evaluation and performance analysis of the proposed multimodal deepfake detection framework.

The image-based, video-based, audio-based, and multimodal fusion models were evaluated independently using benchmark datasets and standard classification metrics. Among the individual modalities, the video-based detection module achieved the highest performance due to its ability to capture temporal inconsistencies across video sequences.

The obtained results validate the effectiveness of modality-specific feature extraction techniques and demonstrate the potential of multimodal learning for deepfake detection applications.

# Chapter 6

# Conclusion and Future Scope

## 6.1 Conclusion

The rapid advancement of deepfake generation technologies has created significant challenges in verifying the authenticity of digital media. Manipulated images, videos, and audio recordings are becoming increasingly realistic, making manual verification difficult and increasing the need for automated detection systems.

This project presented a multimodal deepfake detection system capable of analyzing image, video, and audio data using modality-specific detection pipelines. The image-based module combines EfficientNet-B4 deep visual features with dlib-based facial landmark analysis to identify spatial and geometric inconsistencies. The video-based module focuses on detecting temporal artifacts through frame-level analysis and facial alignment techniques, while the audio-based module utilizes spectrogram-based representations to identify characteristics of synthetic speech.

The extracted feature representations from individual modalities are integrated through feature-level fusion, enabling the system to leverage complementary information from visual, temporal, and acoustic domains. This multimodal approach improves robustness against advanced deepfake generation techniques that may evade single-modality detectors.

Experimental evaluation demonstrated promising performance across all implemented modules. The image-based detection model achieved an accuracy of

86.59%, the video-based detection model achieved 93.02%, and the audio-based detection model achieved 87.59%. The multimodal fusion framework achieved an overall accuracy of 83% by integrating visual, temporal, and acoustic representations into a unified detection pipeline. These results validate the effectiveness of the proposed methodology and highlight the importance of multimodal learning for deepfake detection.

In addition, a graphical user interface was successfully integrated with the framework to support multimedia upload, prediction visualization, and confidence-based analysis for image, video, and audio inputs.

Overall, the developed system provides a scalable and modular solution for multimedia authenticity verification and establishes a strong foundation for future real-time deepfake detection applications.

## 6.2 Achievements of the Project

The major achievements of the proposed project are summarized below:

- Developed an image-based deepfake detection module using EfficientNet-B4 and facial landmark analysis.
- Implemented a video-based deepfake detection pipeline capable of identifying temporal inconsistencies in manipulated videos.
- Developed an audio-based spoof detection module using spectrogram-based deep learning techniques.
- Integrated feature extraction techniques from multiple modalities into a unified multimodal framework.
- Applied feature optimization techniques such as Principal Component Analysis (PCA) to improve computational efficiency.
- Implemented confidence-based prediction and reliability estimation mechanisms.

- Designed a modular architecture that supports independent retraining and future model enhancements.
- Successfully integrated a graphical user interface for multimedia upload and prediction visualization.
- Established a framework suitable for future deployment and real-time multimedia authenticity verification.

## 6.3 Limitations

Although the proposed system demonstrates strong performance, several limitations remain.

The image-based detection module may experience reduced effectiveness when processing highly realistic deepfake images generated using advanced synthesis techniques with minimal visible artifacts.

The video-based module requires processing multiple frames from each video, which increases computational requirements and inference time compared to image-only approaches.

Similarly, the audio-based detection model may face challenges when evaluating synthetic speech generated using state-of-the-art voice cloning systems that closely replicate human speech characteristics.

The current implementation primarily relies on offline inference and further optimization is required for large-scale real-time deployment. Although a graphical user interface has been successfully integrated with the framework, additional backend optimization and deployment scalability improvements are still required for production-level applications.

Finally, performance may vary when evaluated on unseen datasets containing manipulation techniques not represented during training.

## 6.4 Future Scope

Several improvements can be incorporated in future versions of the proposed system.

Future work will focus on improving deployment scalability and real-time inference capability of the implemented graphical user interface. Additional backend optimization and lightweight deployment techniques can further improve usability across practical environments.

Future research can also explore transformer-based architectures and attention-driven fusion strategies to improve multimodal feature integration. Advanced fusion mechanisms may allow the system to better capture relationships between visual, temporal, and acoustic information.

The framework can be extended to support live video stream analysis, enabling continuous monitoring of multimedia content across social media platforms, video conferencing applications, and online communication systems.

Additional modalities such as physiological signals, eye-gaze analysis, lip synchronization verification, and behavioral biometrics may also be incorporated to improve detection robustness.

Deployment through cloud-based services and mobile platforms represents another promising direction for future development. Such deployment would allow large-scale authenticity verification with minimal hardware requirements on the client side.

Furthermore, continual learning techniques can be investigated to ensure that the system remains effective against emerging deepfake generation methods.

## 6.5 Summary

This chapter presented the conclusion of the proposed multimodal deepfake detection system and discussed its major achievements, limitations, and future development opportunities. The results obtained from image, video, audio, and multimodal analysis demonstrate the effectiveness of multimodal learning for detecting manipulated media.

The developed system establishes a strong foundation for future research and deployment in the field of multimedia forensics and digital content authentication. Future improvements in real-time deployment, multimodal fusion optimization, and scalable inference are expected to further enhance the performance and practical applicability of the system.

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